



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Wehunt Preserve; Davidson Homes
Location: Braselton, GA
Description: Hickory - C, Slab

Date: February 9, 2024
By: JTH / TJH
House Faces: Worst East

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,663 ft ²
Volume	23,237 ft ³
Percentage Glass	11 %
Number of Bedrooms	4

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	9761
SEER	14
ACH50	5.0
Mechanical Ventilation	NO
House Outside Air CFM	N/A

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.130	Exterior Door		R-7.7	
	D2	0.083	Garage Door		R-12	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling		R-38 blown	
	R2	0.026	Vaulted		R-38 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.026	Framed		R-38 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.35	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
			Unit 1 - Whole House		
			Cooling	Heating	
Sensible Load			27870 Btu/hr	32477 Btu/hr	
Minimum Nominal Capacity			3.10 Tons	9.52 kW	
Load Density			860 SF/Ton	3.6 W/SF	
Floor Area Served			2663.0 SF	2663.0 SF	
Mechanical Ventilation			N/A CFM	N/A CFM	
System Size			3.5 Tons	15.0 kW	
Leaving Air Temperature			57.2 deg F	103.9 deg F	
Installed Capacity			761 SF/Ton	5.6 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

Project:	Wehunt Preserve; Davidson Homes
Location:	Braselton, GA
Description:	Hickory - C, Slab

Date: February 9, 2024
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House Faces: Worst East

[illegible][illegible]

Unit 1 (Continued)				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs	32	42	37	
Hallway	33	26	29	
Bed 2 WIC	24	34	29	
Bed 2	83	92	87	
Bath 2	18	21	20	
Bed 3	65	57	61	
Bed 3 WIC	10	16	13	
Bed 4 WIC	15	26	21	
Bed 4	80	59	69	
Bath 3	16	19	17	
Owners Bed	176	159	167	
Owners Bath	39	51	45	
Owners WIC	53	69	61	
Laundry	65	28	47	

Cooling Size	1.5	DX Cooling kWh	1838
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By: JTH / TJH

Level(s): First Floor

Best Case	1.5	Total kWh	4871
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SEER 14

COOLING LOAD CALCULATION										Ceiling Height		9		9		9		9		9		9		9		9		9		9		9	
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Cooling Load Factor	Entire House		Foyer & Hall		Study		Powder		Family		Breakfast		Kitchen		Mud		Pantry									
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1	North	Right	(E)	-	6 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	NorthEast		(SE)	-	8 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	East	Front	(S)	-	13 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	SouthEast		(SW)	-	14 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	South	Left	(W)	-	16 Btu/hr-sf	33	521	-	12.5	197	8	126	12.5	197	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	SouthWest		(NW)	-	26 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	West	Back	(N)	-	29 Btu/hr-sf	103	3039	-	-	-	-	-	54	1,593	37.04	1,092	12	354	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	NorthWest		(NE)	-	18 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	Sky Light			-	46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
G1	Shaded			-	6 Btu/hr-sf	48	298	-	12	75	36	224	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Door	D1	Door	21	0.13	2.73 Btu/hr-sf	22	60	12	33	-	-	-	-	-	-	10.01	27	-	-	17.81	44	-	-	-	-	-	-	-	-	-	-	-	
	D2	Door	30	0.08	2.49 Btu/hr-sf	18	44	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Above Grade Walls	W1	North	13	0.08	1.00 Btu/hr-sf	270	270	-	-	20	20	-	-	-	-	-	-	173	173	78	78	-	-	-	-	-	-	-	-	-	-	-	
	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	East	23	0.08	1.77 Btu/hr-sf	123	218	29	52	94	166	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	South	16	0.08	1.23 Btu/hr-sf	367	452	-	-	152	187	55	68	160	197	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	West	23	0.08	1.77 Btu/hr-sf	277	490	-	-	-	-	-	-	128	226	58	103	91	162	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	North	13	0.08	1.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Basement Walls	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	East	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	South	16	0.08	1.23 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	West	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W2	Grade	0	0.07	0.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	Partition	10	0.08	0.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Other	W1	Garage	30	0.08	2.31 Btu/hr-sf	222	513	130	300	-	-	-	-	-	-	-	-	55	128	37	85	-	-	-	-	-	-	-	-	-	-	-	
	W3	Attic	55	0.05	2.92 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W3	Kneewall	55	0.05	2.92 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	R1	Ceiling	55	0.03	1.43 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
Floor	R2	Vaulted	55	0.03	1.43 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	F1	Crawl/Uncon.	11	0.03	0.29 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	F1	Garage	30	0.03	0.78 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	F2	Slab	0	0.10	0.00 Btu/hr-sf	1163	0	153	-	211	-	41	-	313	-	181	-	179	-	57	-	28	-	-	-	-	-	-	-	-	-	-	
Internal Vent	People				255	Btu/hr	2	510	-	1	255	-	1	255	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Lights (W)				3.413	Btu/hr-W	585	1997	77	263	106	362	21	72	157	536	91	311	90	307	29	99	14	48	-	-	-	-	-	-	-		
	Appliances (W)				3.413	Btu/hr-W	930	3174	-	-	-	-	-	250	853	-	680	2,321	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Infiltration	0.25	ACH		0.09	Btu/hr-cf		942		124		171		33		254		147		145		46		23	-	-	-	-	-	-	-	-	
Vent Area	Volume			CF	10467		1377		1899		369		2817		1629		1611		513		252		0		0		0		0		0		
	Area			SF	1163		153		211		41		313		181		179		57		28		0		0		0		0		0		
Subtotal (Btu/hr)							12530		846		1582		299		4111		1680		3461		395		155		0		0		0		0		
Safety Factor							1253																										
Sensible Cooling Load (Btu/hr)							13784																										
Nominal Tons Required (75% sens. cap.)							1.53																										
System Selection 1.5 Tons							600		41		76		14		197		80		166		19		7										
Space CFM							600		41		76		14		197		80		166		19		7										
HEATING LOAD CALCULATION										Ceiling Height		9		9		9		9		9		9		9		9		9		9		9	
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Heating Load Factor	Entire House		Foyer & Hall		Study		Powder		Family		Breakfast		Kitchen		Mud		Pantry		0		0		0		0	
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1		54	0.35	18.9 Btu/hr-sf	184	3478	12	227	48.5	917	8	151	66.5	1,257	37.04	700	12	227	0	-	0	-	0	-	0	-	0	-	0	-	0	
	G2		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
	G3		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-		
Door	D1		54	0.13	7.0 Btu/hr-sf	22	154	12	84	0	-	0	-	0	-	0	-	10.01	70	0	-	0	-	0	-	0	-	0	-	0	-	0	
	D2		54	0.08	4.5 Btu/hr-sf	18	80	0	-	0	-	0	-	0	-	0	-	0	-	17.81	80	0	-	0	-	0	-	0	-	0	-	0	
Walls	W1		54	0.08	4.2 Btu/hr-sf	1037	4313	29.3	122	265.7	1,105	55.3	230	287.7	1,196	57.96	241	263.8	1,097	77.5	322	0	-	0	-	0	-	0	-	0	-	0	
	W2		54	0.07	3.6 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	
	W3		54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	
	W1	Partition	27	0.08	2.1 Btu/hr-sf	222	462	130	270	0	-	0	-	0	-	0	-	0	-	55.49	115	36.7	76	0	-	0	-	0	-	0	-	0	
Floor	W2	Grade	27	0.07	1.8 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	
	R1		54	0.03	1.4 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	
Roof	R2		54	0.03	1.4 Btu/hr-sf	0	0	0																									

SEER 14

COOLING LOAD CALCULATION							Ceiling Height		18		8		8		8		8		8		8		8		8		8		8		8		8				
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Cooling Load Factor	Entire House		Stairs		Hallway		Bed 2 WIC		Bed 2		Bath 2		Bed 3		Bed 3 WIC		Bed 4 WIC		Bed 4		Bath 3		Owners Bed		Owners Bath		Owners WIC		Laundry	
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1	North	Right	(E)	6 Btu/hr-sf	27	166	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	NorthEast		(SE)	8 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	East	Front	(S)	13 Btu/hr-sf	45	587	-	-	-	-	15	196	30	392	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	G1	SouthEast		(SW)	14 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	South	Left	(W)	16 Btu/hr-sf	15	237	-	-	-	-	-	-	-	-	-	15	237	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	SouthWest		(NW)	26 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	West	Back	(N)	29 Btu/hr-sf	60	1770	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	15	442	-	-	45	1,327	-	-	-	-	-	-	
	G1	NorthWest		(NE)	18 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	G1	Sky Light			46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
G1	Shaded			6 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Door	D1	Door	21	0.13	2.73 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	D2	Door	55	0.08	4.57 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Above Grade Walls	W1	North	13	0.08	1.00 Btu/hr-sf	333	334	-	-	-	-	27	27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	137	137	99	99	71	71	-	-	
	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East	23	0.08	1.77 Btu/hr-sf	213	377	-	-	-	-	32	57	87	154	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	65	115	28	50		
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South	16	0.08	1.23 Btu/hr-sf	344	424	-	-	-	-	-	-	-	104	128	48	59	88	108	52	64	53	66	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West	23	0.08	1.77 Btu/hr-sf	291	515	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	49	86	98	174	50	88	95	167	-	-	-	-		
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	North	13	0.08	1.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Basement Walls	W1	NorthEast	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	East	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthEast	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	South	16	0.08	1.23 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	SouthWest	21	0.08	1.62 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	West	23	0.08	1.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	NorthWest	19	0.08	1.46 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W2	Grade	0	0.07	0.00 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W1	Partition	10	0.08	0.77 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
Other	W1	Garage	30	0.08	2.31 Btu/hr-sf	93	214	93	214	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W3	Attic	55	0.05	2.92 Btu/hr-sf	93	272	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	W3	Kneewall	55	0.05	2.92 Btu/hr-sf	49	141	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	R1	Ceiling	55	0.03	1.43 Btu/hr-sf	1499	2144	76	109	154	220	40	57	148	212	69	99	147	210	31	44	32	46	147	210	51	73	268	383	138	197	120	172	78	112		
	R2	Vaulted	55	0.03	1.43 Btu/hr-sf	0	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Canilever	20	0.03	0.52 Btu/hr-sf	11	6	-	-	-	-	11	6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F1	Garage	30	0.03	0.78 Btu/hr-sf	253	197	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	F2	Slab	0	0.10	0.00 Btu/hr-sf	77	0	77	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-			
	Internal	People				255 Btu/hr	4	1020	-	-	-	-	-	1	255	-	1	255	-	-	-	1	255	-	-	-	1	255	-	-	-	60	205	39	133		
Lights (W)					3.413 Btu/hr-W	753	2570	39	133	77	263	20	68	74	253	35	119	7	253	16	55	16	55	74	253	26	89	134	457	69	235	60	205	39	133		
Appliances (W)					3.413 Btu/hr-W	200	683	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	200	683	-	-		
Infiltration		0.25	ACH		0.09 Btu/hr-cf	1149		125		111		29		107		50		106		22		23		106		37		193		99		86		56			
Vent	Volume			CF	12770		1386		1232		320		1184		552		1176		248		256		1176		408		2144		1104		960		624				
	Area			SF	1500		77		154		40		148		69		147		31		32		147		51		268		138		120		78				
Subtotal (Btu/hr)							12806		580		594		440		1499		327		1169		185		275		1440		286		3178		699		956		1178		
Safety Factor							1281																														
Sensible Cooling Load (Btu/hr)							14087																														
Nominal Tons Required (75% sens. cap.)							1.57																														
System Selection 2 Tons Space CFM							800		36		37		27		94		20		73		12		17		90		18		199		44		60		74		
HEATING LOAD CALCULATION																																					
Components					CLTD (°F)	U-Value (Btu/hr-sq-F)	Heating Load Factor	Entire House		Stairs		Hallway		Bed 2 WIC		Bed 2		Bath 2		Bed 3		Bed 3 WIC		Bed 4 WIC		Bed 4		Bath 3		Owners Bed		Owners Bath		Owners WIC		Laundry	
	Type	Description		Qty				Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty
Glass	G1			54	0.35	18.9 Btu/hr-sf	147	2774	0	-	0	-	15	284	30	567	0	-	0	-	15	284	0	-	15	284	0	-	63.75	1,205	4	76	4	76	0	-	
	G2			54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	G3			54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
Door	D1			54	0.13	7.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
	D2			54	0.08	4.5 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
Walls	W1			54	0.08	4.2 Btu/hr-sf	1181	4911	0	-	0	-	59.25	246	190.5	792	47.97	199	87.78	365	51.75	215	102.1	424	98.22	408	49.5	206	231	960	98.78	411	136	565	28.49	118	
	W2</																																				